RISHIRAJ NAGARAJAN

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA — Expected Dec 2026
Master of Computational Data Science
Coursework: Deep Learning, Large Language Models and their Applications, Large Language Model Systems, Cloud Computing, Machine Learning, Foundations of Computational Data Science

Vellore Institute of Technology, Vellore, India — Jun 2025
Bachelor of Technology, Computer Science Engineering
Coursework: Design and Analysis of Algorithms, Artificial Intelligence, Big Data Analytics

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, CUDA
ML & LLMs: PyTorch, HuggingFace, vLLM, SGLang, LangChain, RAG, LoRA/PEFT, scikit-learn, XGBoost, Anthropic/OpenAI/Gemini APIs
Infrastructure & DevOps: AWS, GCP, Docker, Kubernetes, Apache Temporal, FastAPI, Spring Boot, Postgres, MongoDB, Redis, Git, CI/CD

EXPERIENCE

Different Labs (General Catalyst-backed) | *ML Systems Engineer Intern* — May 2026 to Present
- Built an applied-LLM edit-memory system that mines styling preferences from human v1 to vMax edit diffs and reinjects them at generation time, gated by a deterministic verifier that re-checks each rule's cited evidence verbatim against the source diff; validated with a non-inferiority eval harness on edit-distance-to-final.
- Engineered a weekly per-organization Apache Temporal pipeline pairing a Java/Spring Boot orchestrator with a Python/FastAPI LangChain ReAct agent whose tool-use loop proposes, reinforces, and retires preference rules via support-count voting and contradiction control.
- Deployed a multi-platform (Meta, Google, LinkedIn) brand-intelligence agent to production on Railway (FastAPI, Redis/ARQ, Postgres; multi-tenant, CI/CD), profiling brand voice from ~150 scraped ads per run via Gemini vision.

Carnegie Mellon University, Language Technologies Institute | *Graduate Researcher* — Apr 2026 to Present
- First author of "Personalized Sycophancy: How Conversational Context Causes LLMs to Abandon Honest Social Judgments" (in preparation; results available on request), advised by Prof. Maarten Sap.
- Constructed SOCIALDRIFT-100, a 100-scenario benchmark quantifying sycophantic drift across multi-turn advice conversations, and identified the self-justification turn as the point most strongly associated with that drift.

Carnegie Mellon University | *Teaching Assistant* — Apr 2026 to Present
- Support Mathematical Foundations for Computational Data Science and AI Technicians, an applied-AI and cloud-native upskilling program delivered to U.S. Army officials.

Brandidea Granular Analytics | *Data Science Intern* — Jan 2025 to May 2025
- Trained and benchmarked Random Forest and XGBoost models for national cement demand forecasting; XGBoost achieved 15% lower RMSE than the incumbent analyst-built Excel baseline.
- Replaced a legacy Excel workflow with an end-to-end ML pipeline and an executive Power BI dashboard (district-level choropleth of India), built on a seasonality-aligned feature pipeline from self-built web scrapers with SMOTE for class imbalance; presented findings directly to the CEO.

SELECTED PROJECTS

Structure-Aware KV Cache Eviction for LLM Serving | SGLang, CUDA, PyTorch — Jan 2026 to Present
- Co-developed two lossless structure-aware prefix-cache eviction policies (Weighted LFU-LRU and Weighted Segmented LRU) in SGLang's RadixAttention, after prototyping an attention-aware variant with custom CUDA kernels in vLLM; on a shared-prefix workload (Qwen2.5-7B, A100), cut P99 time-to-first-token 39.6% and mean TTFT 26% against the LRU baseline with throughput within 1%. Submitted upstream as an SGLang PR (pending review).

Jump Trading Poker AI, Competition Finalist | Python, PyTorch, Monte Carlo, NFSP — Mar 2026
- Reached finalist standing by building a 27-card heads-up poker AI with a preflop equity table over C(27,5) = 80,730 hands precomputed by Monte Carlo for O(1) lookup, exact postflop equity enumeration, and a Nash-seeking NFSP agent trained via reservoir-sampled supervised learning.

HARBOR: Behavioral Health Risk Assessment | XGBoost, scikit-learn, PyTorch — Jan 2026 to Present
- Built an end-to-end ML pipeline predicting depression severity from 50+ smartphone-sensor features (218 students, 35K+ observations); patient-based cross-validation with zero participant overlap exposed a generalization failure in published behavioral-ML benchmarks, then LoRA-fine-tuned MedGemma-27B on a single A100 against that honest baseline, a documented negative result.

Self-Healing Agent | Anthropic API, AWS CloudWatch, Slack — Dec 2025
- Architected an autonomous system that monitors AWS CloudWatch, deduplicates stack traces via deterministic fingerprinting, and routes them through a two-stage pipeline (Claude Haiku triages noise; Claude Sonnet generates a diff patch plus pytest suite) to produce a fix proposal in roughly two minutes.
- Built a human-in-the-loop Slack approval flow (Block Kit) for concurrent fix proposals; on approval the agent branches off main, applies the patch, and opens a GitHub PR, with errors tracked in Supabase for recurrence detection.